

18.06 Pset 2

Due Monday September 26 (since Friday Sep 23 is a holiday) at 11am on GradeScope.
Submit PDF of your handwritten solutions along with your (clearly labeled) Julia code.

Problem 1 (7+3+5 points)

Consider the following 5×5 matrix, which is *almost* upper triangular:

$$A = \begin{pmatrix} 2 & -2 & 0 & 1 & 1 \\ 2 & -1 & 1 & -2 & 1 \\ & 3 & 4 & -3 & 1 \\ & & -2 & -5 & -3 \\ & & & 7 & 3 \end{pmatrix}$$

Corresponding in Julia to:

```
In [ ]: A = [2 -2 0 1 1; 2 -1 1 -2 1; 0 3 4 -3 1; 0 0 -2 -5 -3; 0 0 0 7 3]
```

(a)

Compute (by hand) its LU factorization $A = LU$ via Gaussian elimination (i.e. give both L and U).

Notice any special pattern to the nonzero entries in L and/or U ?

(b)

Why doesn't your hand calculation match the output of Julia's `lu(A)` function? Try it, it's not even close! (Do `using LinearAlgebra` in order to access the `lu` function.)

(c)

Suppose you carried out arithmetic at the *same rate*, but A was **10 times** larger: a 50×50 nearly upper triangular matrix (upper triangular + nonzeros just below the diagonal).

About how much longer should part (a) take? 10 times, 100 times, 1000 times?

Is this different from how it would scale for a general matrix with no special pattern of zero entries?

Problem 2 (8+2 points)

Consider the three 4×4 matrices

$$A = \begin{pmatrix} 1 & -3 & 0 & 3 \\ & 1 & 1 & 2 \\ & & 1 & -1 \\ & & & 1 \end{pmatrix}, B = \begin{pmatrix} -3 & -1 & 1 & -2 \\ -3 & 0 & 0 & -3 \\ 1 & 3 & 3 & -3 \\ 2 & 3 & 2 & 0 \end{pmatrix}, C = A(AB)^{-1}$$

(a) Compute the third column of C^{-1} without computing the whole inverse of any matrix. Think before plugging-and-chugging!

(b) Check your answer by explicitly computing C^{-1} in Julia.

```
In [ ]: # fill these in:
A = ??????
B = ??????

C = A / (A*B) # computes A(AB)^{-1}
C^-1 # computes C^{-1}
```

Problem 3 (12 points)

(Based on Strang problems.)

Which of the following subsets of $\mathbb{R}^3$ are actually subspaces? (Recall that $[x_1, x_2, x_3]$, with commas, denotes a column vector.) If it is *not* a subspace, give a rule that it violates.

1. The plane of vectors $[x_1, x_2, x_3]$ with $x_1 = x_2$
2. The plane of vectors $[x_1, x_2, x_3]$ with $x_1 = 1$
3. The vectors $[x_1, x_2, x_3]$ satisfying $x_1 x_2 x_3 = 0$
4. All linear combinations of $v = [1, 4, 0]$ and $w = [2, 2, 2]$
5. The vectors $[x_1, x_2, x_3]$ satisfying $x_1 + x_2 + x_3 = 0$
6. The vectors $[x_1, x_2, x_3]$ satisfying $x_1 \leq x_2 \leq x_3$

Problem 4 (10 points)

Nowadays, numerical linear algebra programs deal with small pivots by swapping rows, leading to the $PA = LU$ factorization. Prof. Edelman mentioned that, in the elder days, people would occasionally swap *columns* instead, leading to a factorization $AP = LU$ for some (different) permutation P (and different L and U matrices).

Given a factorization $AP = LU$ for a nonsingular matrix A , outline how you would use it to solve $Ax = b$ for x (analogous to how we outlined the use of $PA = LU$ in class).

(You can exploit the fact, explained in section 2.7 of the textbook, that you can invert permutations "for free": $P^{-1} = P^T$ for any permutation matrix P , where the transpose P^T of a matrix is simply formed by swapping the rows of P with the columns of P . We will spend more time on this later in 18.06.)

Problem 5 (9 points)

Suppose we have a 3×3 matrix A . We will transform this into a *new* matrix B by doing operations on the rows or columns of A as follows. For each part, explain how to compute B as $B = EA$ or $B = AE$ (say which!) and give the matrix E .

1. Reverse the order of the rows of A .
2. Add the second row to the third row, *then* replace the second row with the sum of the first and (new) third rows.
3. Subtract the third column from the first and second columns.